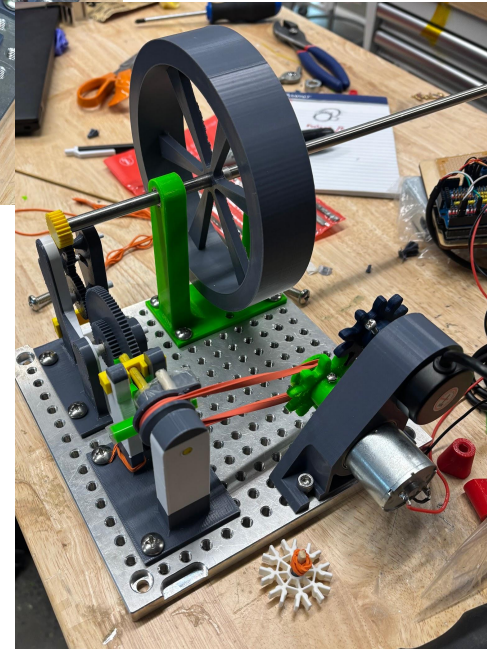
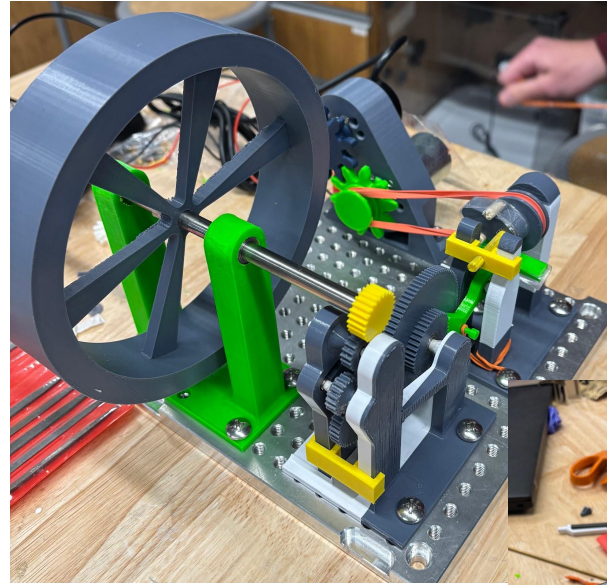


“Homie” Rekt Analysis

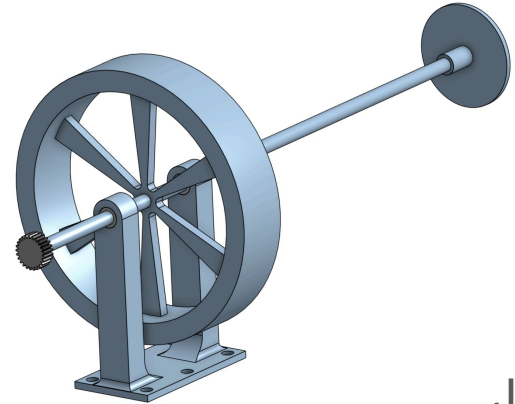
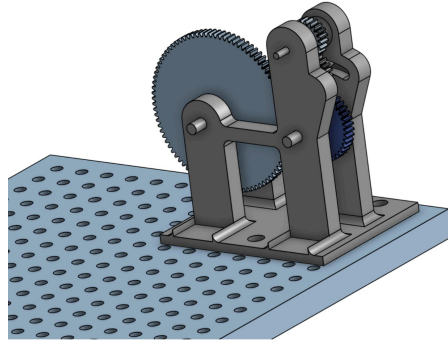
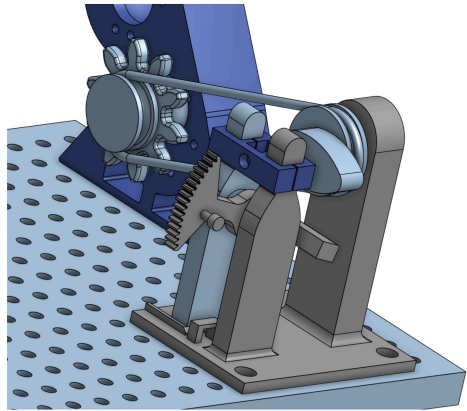
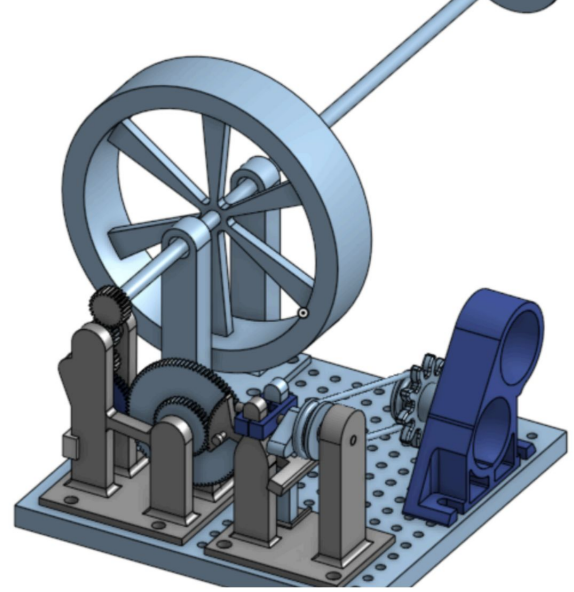
By: Justin Pai, Aleks Szetela



Homie Rekt

Purpose of design was to keep an output shaft at a constant angular velocity

Made up of multiple subsystems to achieve this effect



Key Assumptions

Sliding gear acts as a normal gear

Rubber bands were perfectly rigid

The torque transferred from the cam to the lever was approximated using the ratio of their radii

Moment of Inertia and damping constant for system were combined

Theoretical Model

Motor and pulley:

$$V(t) - I(t)R - L\frac{dI(t)}{dt} = K_b\omega_m(t)$$

$$V(s) - I(s)R - LsI(s) = K_b\Omega_m(s)$$

$$I(s) = \frac{V(s) - K_b\Omega_m(s)}{Ls + R}$$

$$J_m\dot{\omega}_m(t) + b_m\omega_m(t) = K_m I(t) - \tau_{cam}(t)$$

$$(J_ms + b_m)\Omega_m(s) = K_m I(s) - T_{cam}(s)$$

Cam and Lever:

$$J_{lever}\dot{\omega}_{lever}(t) + b_{lever}\omega_{lever}(t) = \frac{1}{N_L}\tau_{cam}(t) - \tau_{lever_out}(t)$$

$$(J_{lever}s + b_{lever})\Omega_{lever}(s) = \frac{1}{N_L}T_{cam}(s) - T_{lever_out}(s)$$

Sliding Gear:

$$J_{sg}\dot{\omega}_{sg_out}(t) + b_{sg}\omega_{sg_out}(t) = \frac{1}{N_{sg}}\tau_{lever_out}(t) - \tau_{mesh}(t)$$

$$(J_{sg}s + b_{sg})\Omega_{sg_out}(s) = \frac{1}{N_{sg}}T_{lever_out}(s) - T_{mesh}(s)$$

Theoretical Model Cont.

Flywheel and Output shaft:

$$J_f \dot{\omega}_f(t) + b_f \omega_f(t) = \frac{1}{N_{final}} \tau_{mesh}(t)$$
$$(J_f s + b_f) \Omega_f(s) = \frac{1}{N_{final}} T_{mesh}(s)$$

Speed Relation:

$$\Omega_{lever}(s) = N_L \Omega_m(s)$$
$$\Omega_{sg_out}(s) = (N_{sg} N_L) \Omega_m(s)$$
$$\Omega_f(s) = (N_{final} N_{sg} N_L) \Omega_m(s) = N_{total} \Omega_m(s)$$

Transfer Function:

$$G(s) = \frac{\Omega_f(s)}{V(s)} = \frac{N_{total} K_m}{(Ls + R)(J_{eq}s + b_{eq}) + K_m K_b}$$

Parameters

$$J_{eq} = J_m + J_{lever}N_L^2 + J_{sg}(N_{sg}N_L)^2 + J_f(N_{total})^2$$

$$b_{eq} = b_m + b_{lever}N_L^2 + b_{sg}(N_{sg}N_L)^2 + b_f(N_{total})^2$$

$$N_{total} = 3.4488$$

$$J_{eq} = 0.0968 \text{ kg} \cdot \text{m}^2$$

$$b_{eq} = 0.0936 \text{ N} \cdot \text{m}/(\text{rad/s})$$

Motor Specs:

19:1 Metal Gearmotor 37Dx52L mm 12V (Helical Pinion)



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Price break	Unit price (US\$)
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5	35.83
25	32.97
100	30.33

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No-load speed @ 12V: 530 rpm

No-load current @ 12V: 0.2 A

Stall current @ 12V: 5.5 A³

Stall torque @ 12V: 8.5 kg·cm³

Max output power @ 12V: 12 W

$$K_b = \frac{V - I_{nl}R}{\omega_{nl}}$$

$$K_b = \frac{12 - (0.2 \times 2.18)}{55.5} \approx \mathbf{0.208 \text{ V} \cdot \text{s}/\text{rad}}$$

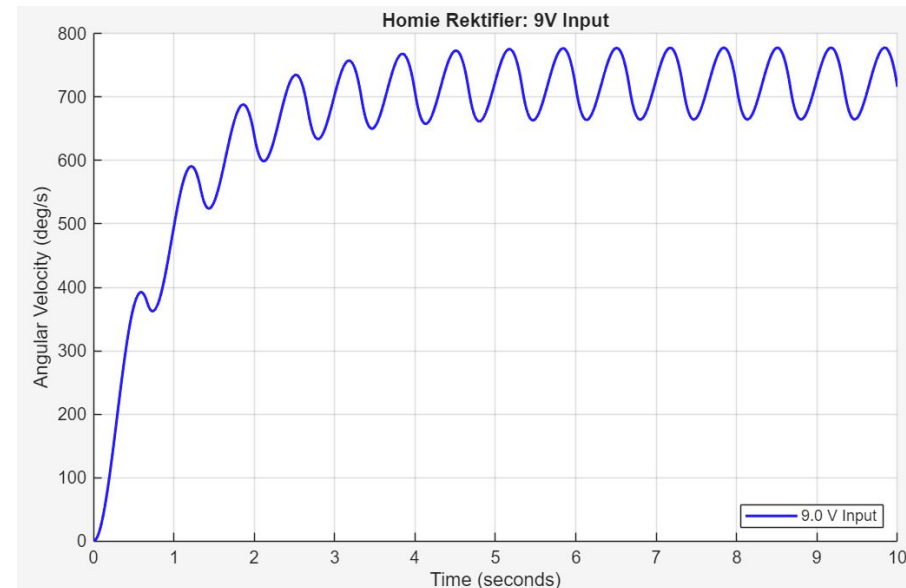
$$K_m = \frac{T_{stall}}{I_{stall} - I_{nl}}$$

$$K_m = \frac{0.834}{5.5 - 0.2} \approx \mathbf{0.157 \text{ Nm}/\text{A}}$$

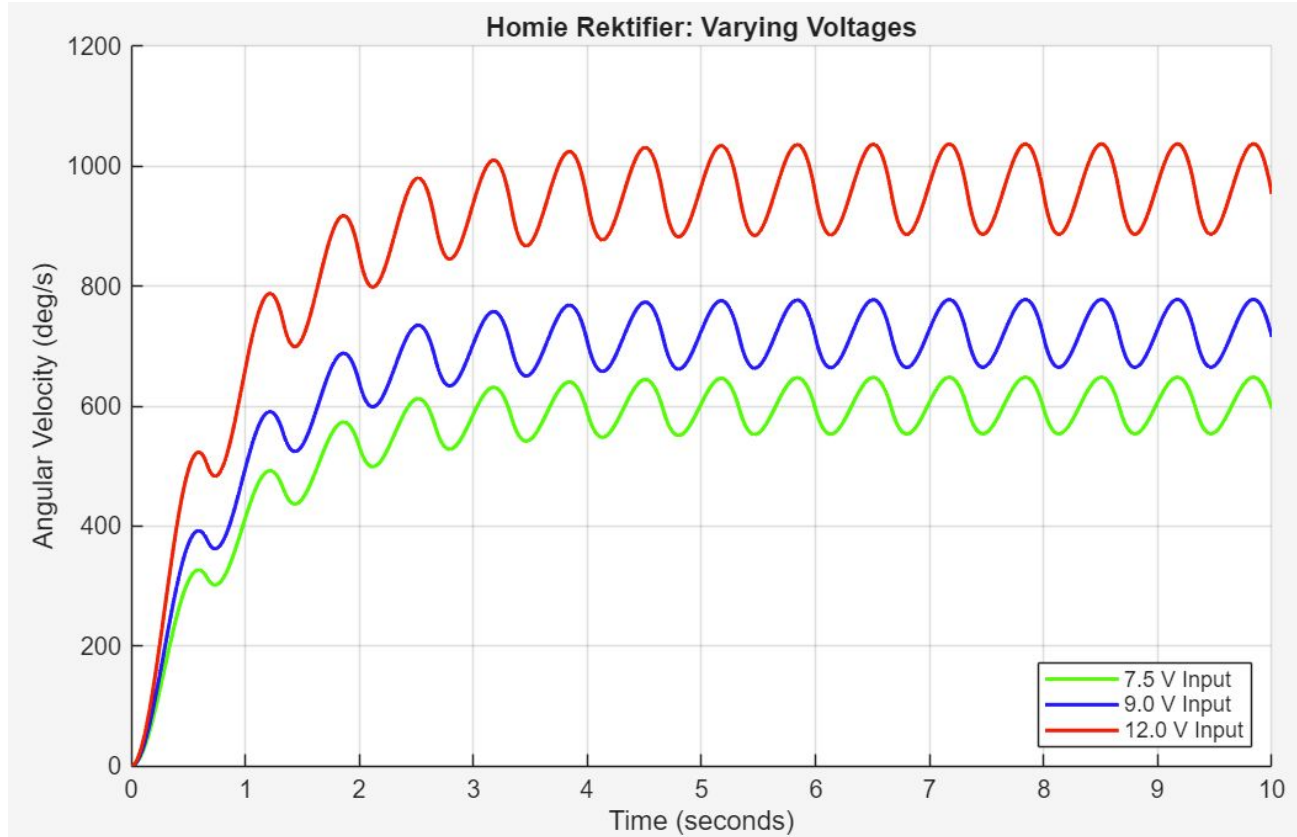
Transient Properties

$$G(s) = \frac{2339.2}{s^2 + 948.79s + 1063.1}$$

- Natural Frequency (ω_n): $\sqrt{1063.1} = 32.6$ rad/s
- Damping Ratio (ζ): $\frac{948.79}{2(32.6)} = 14.55$
- Dominant Time Constant (τ): 0.892 s
- Rise Time ($t_r \approx 2.2\tau$): 1.96 s
- Settling Time ($t_s \approx 4\tau$): 3.57 s
- Percent Overshoot (%OS): 0%



Varying Input Voltage



Varying Gear Train Ratio

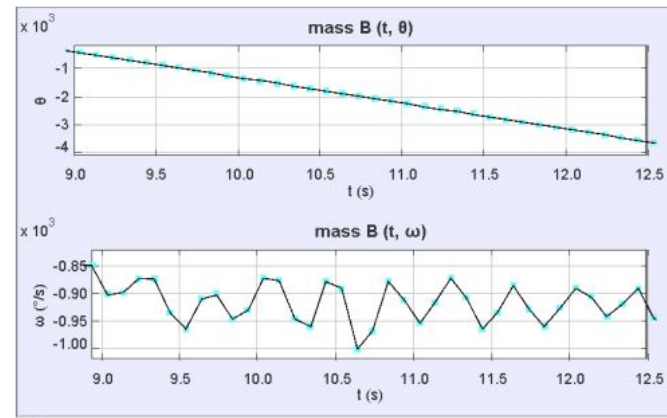
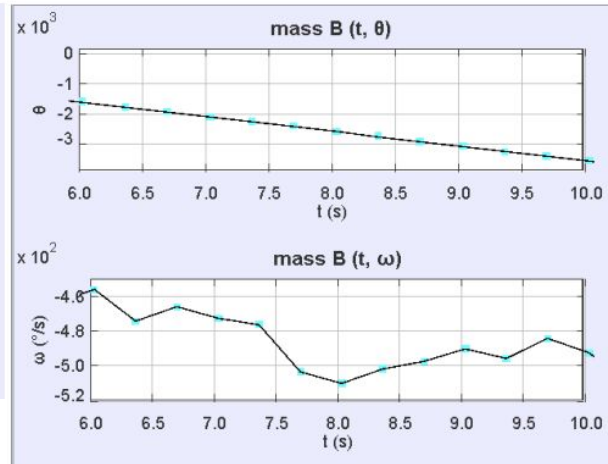
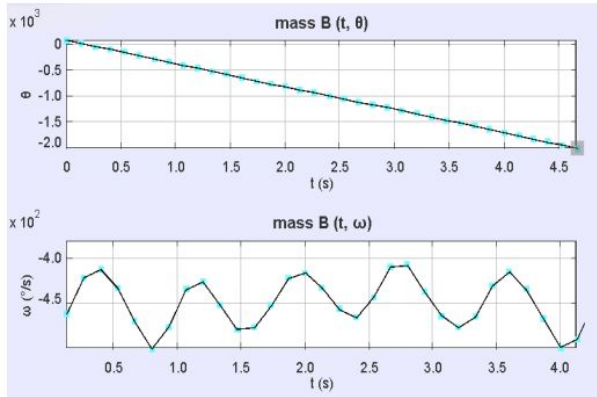


Experimental Results

7.5V

9V

12V



Voltage	Theoretical Rectified ω range ($^\circ/\text{s}$)	Experimental Rectified ω range ($^\circ/\text{s}$)
7.5 V	553-646	420-480
9 V	664-774	480-510
12 V	886-1036	850-970

Future Work

- Add more subsystems. Ex: Centrifugal Governor
- Optimize gear ratio and inertia of flywheel
- Switch rubber band pulley for a more efficient method of transferring torque

A special thank you to Mike Giglia for giving us the motor specs and assigning the Mechanical Rectifier project in EID 103